

Blazor applications

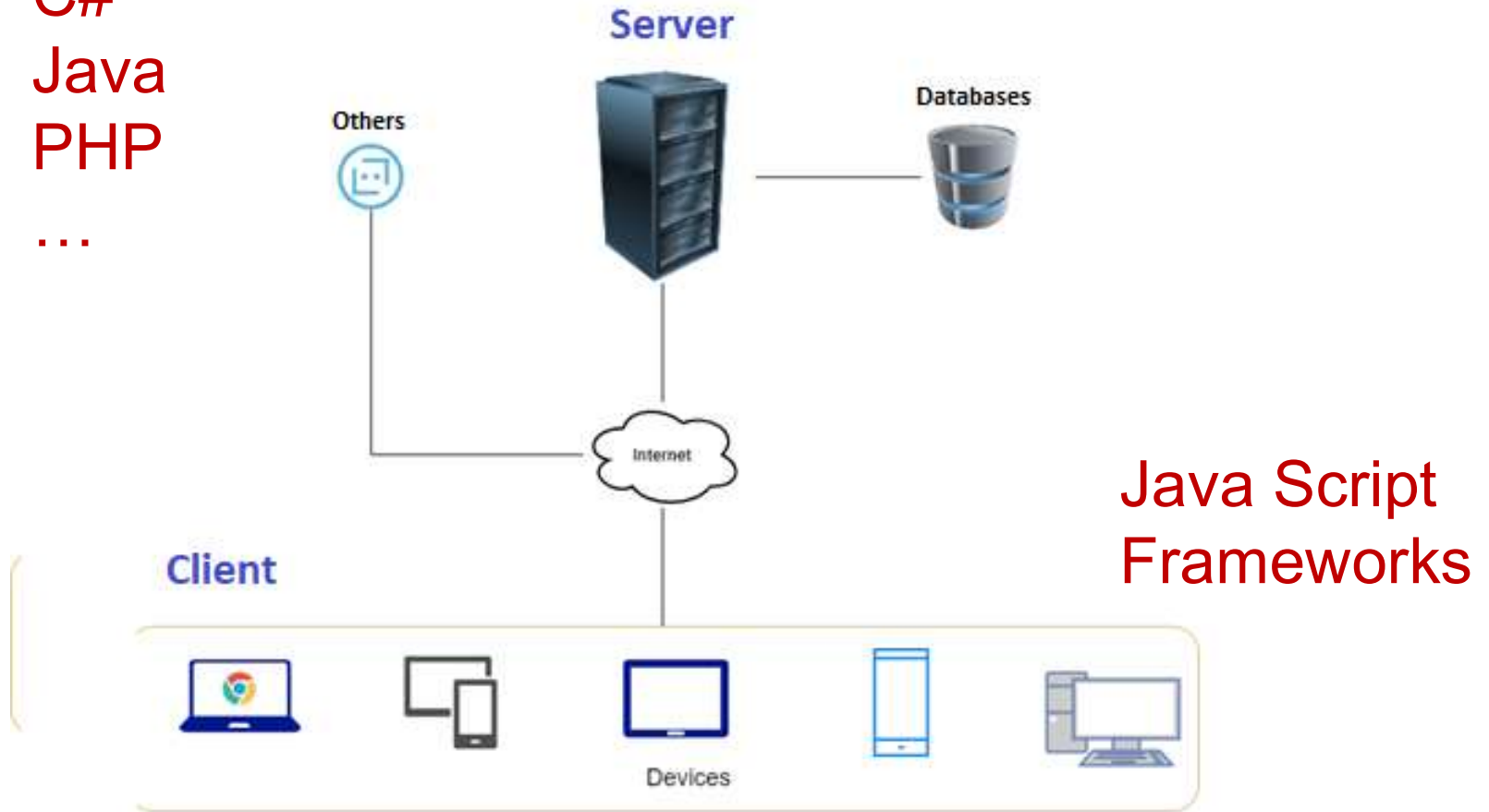
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Agenda

- Application development
- ASP.NET Core Razor components
- Summary

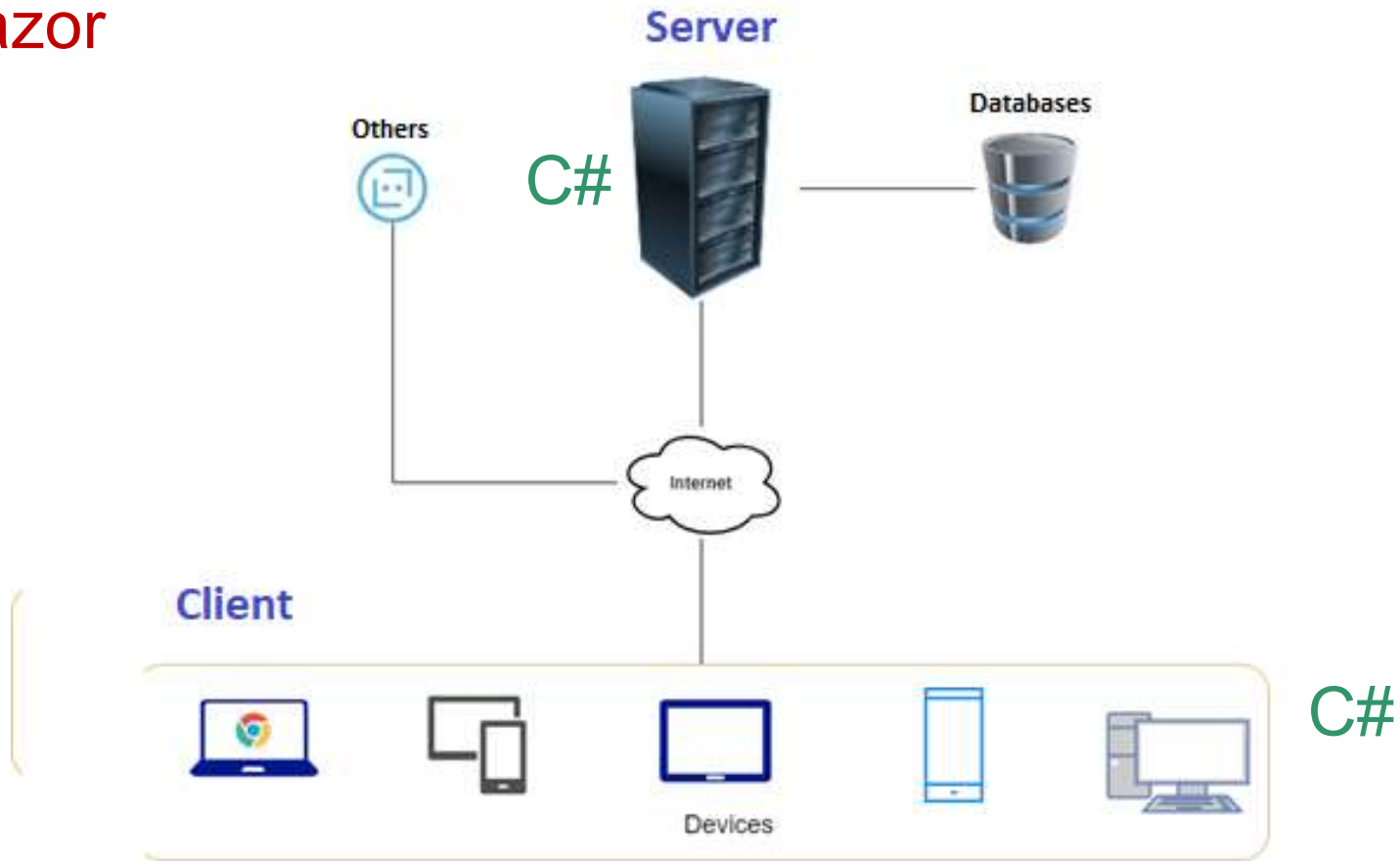
How we develop applications

C#
Java
PHP
...

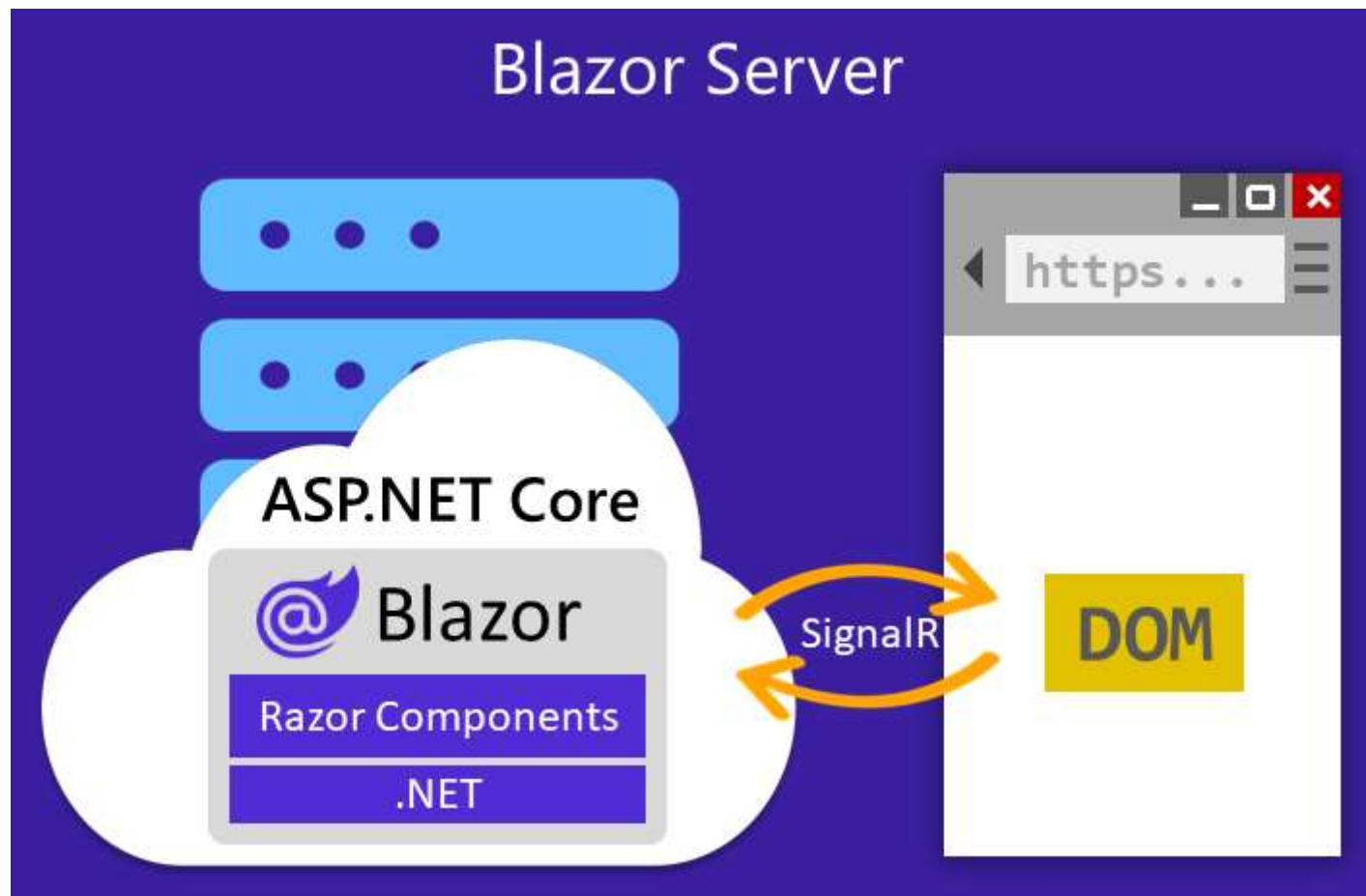


C# for both server and client sides

Blazor



Blazor server



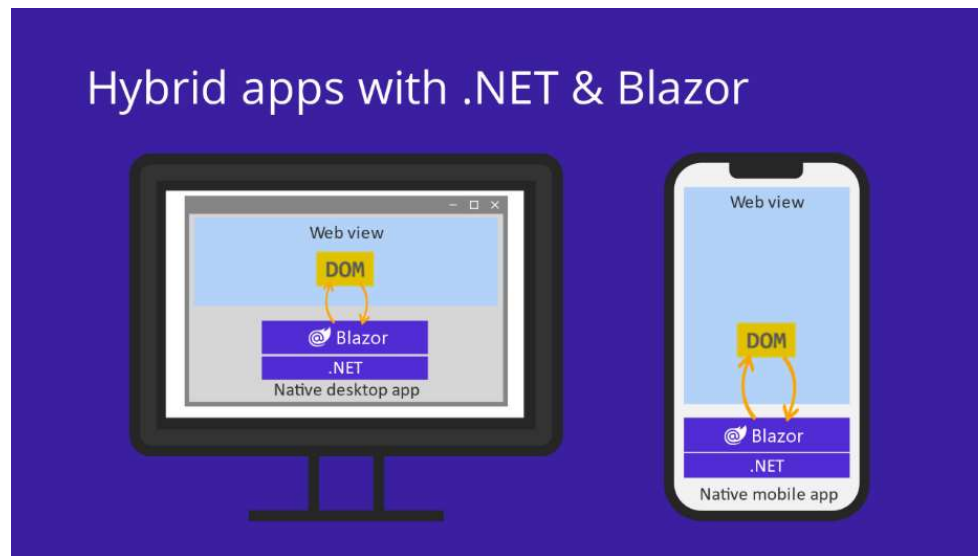
Blazor assembly



Web assembly apps

- An app exclusively runs on the Blazor WebAssembly hosting model without server-side rendering and interactivity → standalone Blazor WebAssembly app
- Standalone Blazor WebAssembly app + backend ASP.NET Core → hosted BlazorWebAssembly app
 - Via a variety of messaging frameworks and protocols: WEB API, gRPC-web, ...

Blazor hybrid



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Razor components

- Blazor apps are built using Razor components
- Component: self-contained portion of user interface (UI) with processing logic to enable dynamic behavior
- Can be nested, reused, shared among projects
- Blazor ~ component driver framework

Example

```
@page "/counter"
<h1>Counter</h1>
<p>Current count: @currentCount</p>
<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>
@code
{
    private int currentCount = 0;
    private void IncrementCount()
    {
        currentCount++;
    }
}
```

Split components

- Split HTML and C# codes
- 2 approaches
 - Partial files
 - Base classes

Partial files

```
// Counter.razor
```

```
@page "/counter"
```

```
<h1>Counter</h1>
```

```
<p>Current count: @currentCount</p>
```

```
<button class="btn btn-primary" @onclick="IncrementCount">Click me</button>
```

```
// Counter.razor.cs
```

```
public partial class Counter
```

```
{
```

```
    private int currentCount = 0;
```

```
    private void IncrementCount()
```

```
    {
```

```
        currentCount++;
```

```
    }
```

```
}
```

Base classes

```
1 @page "/counter"
2 @inherits CounterBase
3
4 <PageTitle>Counter</PageTitle>
5
6 <h1>Counter</h1>
7
8 <p role="status">Current count: @currentCount</p>
9
10 <button class="btn btn-primary" @onclick="IncrementCount">Click me</button>
```

```
1 namespace FirstApp.Pages;
2 protected class CounterBase: ComponentBase{
3     private int currentCount = 0;
4
5     private void IncrementCount()
6     {
7         currentCount++;
8     }
9 }
```

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Summary

- Blazor allows using C# in both server and client sides
- Blazor apps are built using Razor components
- Razor components can be either in a single file, partial files or base class

Q&A
